Class Test Unit-1

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25/25 Good

Q1) You are designing a hypothesis space H for wheather prediction with 5 nominal attributes (Temp{Hot, mild, cold}. If each attribute has 3 possiable values and the hypothesis allows ? as a wild card How many possiable hypotheses are in H?

Sol

5 Nominal attribute

3 Possiable values.

$H = (d+1)^n$

d = 3 – No of values per attribute

n = 5 – No of attribute.

1 – represent the wild card.

$H = (3+1)^5$

$H = 4^5 = 1024$

$H = 4^5$

2) Design a version space of a binary classification Problem with two attributes A & B. How many hypotheses will exist in H if each attribute can be 0, 1, ?

Two Attribute A & B.

$H = K^n$

K = 3 & n = 2

K = choices.

$H = 3^2 = 3 \times 3 = 9$

$H = 3^2$

youre implementing the candidate elimination algorithm in a medical diagnosis system. After receiving conflicting positive and negative dist what should the learner do to maintain consi

sol candidate Elimination maintain two boundaries.

S: most specific consistent HYpotheses.

G: most general consistent hypotheses.

When Negative.

Prune G to remove inconsistent hypotheses and Generalizes.

to a sing

4a) If the version space is reduse. hypothesis what can be infassed.

sol

$V = G \cap S$

$|V| = 1$

The-Target has been precisely learned.

Q5) You're tasked with designing an algorith to learn a disjunctive Normal form(DN concept. which of the following modifica helps reduse the size of hypothesis.

sol

DNF means Disjunctive Normal form

* OR operations btw that may contain AND oper tion.

EX:

$(A \wedge B) \vee (C \wedge D)$

(A) Restrict to a single disjunction.

sol

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4a) If the version space is reduce. hypothesis what can be inferred.

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Q6) A robot learns to Navigate using candidate Elimination. If the environment contain random obstacles (Noise). best strategy to handle incorrect

sol

Noise means that some training examples may be incorrect or inconsistent with the true concept.

"probabilistic method ranks hypotheses" by their likelihood or probability instead.

Q7) You're building a classifier for rare disease. The training set is highly imbalanced design strategy in concept learning best.

sol

$$E_w = \sum_{i=1}^{N} w_i I(h(x_i) \neq y_i)$$

w_i = weight assigned to ex

I = 1 Prediction.

N = total No. of ex

use weighted hypothesis scoring

Q8) In a hypothesis space with 6 binary attr each attribute allowing values {0, 1, ?} What is the total Number of possible hypot

sol

$n = 6$.

$$H = 3^n - 1$$

$H = 3^6 - 1$

$H = 729 - 1 = 728$

$$H = 3^6 - 1$$

Q9) A version space start with G = {(?, ?, ?)} a $S = \{(\perp, \perp, \perp)\}$ Given 4+ and 3⊖ instance assuming each instance pruned half th hypotheses in G. How many hypotheses.

Sol 4 Positive & 3 Negative

Go then after m eleminations

$$G_m = G_0(1/2)^m$$

$$G_7 = G_0(1/2)^7 = \frac{G_0}{128}.$$

The Question is incomplete. option B(4), given in the PDF, would required additional assumption.

Q10) In an inductive reasoning task with 1000, instance and a hypothesis space of 10,000 hypotheses VC dimension upper bound.

Sol

Given,

1000 instances & 10,000 hypotheses

$$d \approx \log_2 |H|$$

d = dimension.

$$d = \log_2(10000)$$

since $2^{13} = 8192$

$2^{14} = 16384$

$\log_2 |H|$

$$d \approx 13.29$$

Q11). Given 3 binary attributes, what is the total Number of hypothesis in the hypothesis space considering generalization symbol?

sol 3 attributes.

$H = 3^n$

$H = 3^3 = 3 \times 3 \times 3 = 27$

$H = 3^3$

Q12) A learner has a hypothesis space of size 256. If each training example eliminates 25% of inconsistent hypotheses. How many examples are needed to reduce.

sol $H_0 = 256$ (initial hypotheses)
$H_n = 16$ (final hypotheses)

$H_n = H_0 (1 - 0.25)^n$

$16 = 256 (0.75)^n$

$\frac{16}{256} = (0.75)^n$

Taking log.

$n = \frac{\log(16/256)}{\log(0.75)}$

$n = \frac{\log(0.0625)}{\log(0.75)} \approx 9.64$

$n = 10$

Q13) In the candidate elimination Algorithm many minimal specialization are generated from the most general hypothesis $(?, ?, ?)$ when a Negative example (sunny, warm, High).

Sol

G = (?, ?, ?)

x = (Skinny, warm, high)

3-attributes.

3-1 = 2

N = 3(3-1)

N = 3 × 2 = 6

N = 6

Q14) Given that the version space V = G ∩ S, and G contain 3 General hypotheses while S contain 4 Specific ones. What is the maximum No of hypotheses.

Sol

|G| = 3

|S| = 4

Version (V) = G ∩ S

$|V| \leq \min(|G|, |S|)$

$\therefore |V| \leq \min(3, 4) = 3$

Unknown without domain.

15) The size of hypothesis space with n ternary attribute allowing '?' generalization is.

Sol

3 attributes.

$H = (3+1)^n$

$H = 4^n$.

Q16) Find-algorithm after receiving (4) examples (Red, small), (Red, ?) what will be the current hypothesis.

Sol Given

x_1 = (Red, small)

x_2 = (Red, ?)

S-algorithm

S_1 = (Red, small)

S_2 = (Red, ?)

G_2 = (Red, ?)

Q17) A learner start with G = {(?,?)} A Negative instance (Sunny, High) eliminate general hypothesis. Each attribute has 2 values.

Sol

G = (?, ?)

Negative instance

x = (Sunny, High)

2-attributes.

h_1 = (Not Sunny, ?)

h_2 = (?, Not High)

N = 2(2-1) = 2

N=2

Q18) A Dataset has 80(+), 20(-) examples. Calculated entropy of dataset

Sol

(+) = 80

(-) = 20

Total examples = 100

Calculate Probabilities

$$P+ = \frac{80}{100} = 0.8$$

$$P- = \frac{20}{100} = 0.2$$

Formula

$$H(S) = -P+ \log_2 P+ - P- \log_2 P-$$

$$H(S) = -0.8 \log_2 (0.8) - 0.2 \log_2 (0.2)$$

$$\log_2 (0.8) \approx -0.322$$

$$\log_2 (0.2) \approx -2.322$$

$$H(S) = -(0.8)(-0.322) - (0.2)(-2.322)$$

$$H(S) = 0.2576 + 0.4644$$

$$\boxed{H(S) \approx 0.722 \text{ bits}}$$

Q19) In a decision tree you are evaluating a split on attribute A. The Parent Node has entropy 1.0, and the split result in two child Nodes of size 40 & 80 Entropies 0.8 what information gain.

sol

Parent entropy = 1.0

Left child: 40 sample, entropy = 0.8

Right child: 60 sample entropy = 0.6

Total sample = 100

Formula

$$IG = H(S) - \sum_i \frac{|S_i|}{|S|} H(S_i)$$

Calculate weighted child entropy

$$H_{children} = \frac{4}{100}(0.8) + \frac{60}{100}(0.6)$$

$$= 0.32 + 0.36 = 0.68$$

$$IG = 1.0 - 0.68$$

$$\boxed{IG = 0.32}$$

Q20) Consider a decision tree constructed usi ID3 algorithm. If an attribute has zer information.

sol

$$\boxed{IG = H(S) - H(S/A)}$$

$$IG = 0$$

$$H(S) = H(S/A)$$

This means splitting on Attribute A does n4

It can't be used for splitting.

Q21) For a binary classification, a decision tree with max depth 4 classified a dataset perfectly with 16 leaves.

Sol

A binary decision tree has 16 leaves.

$$L = I + 1$$

L = No. of leaves
I = No. of internal decision Nodes.
internal

$$I = 16 - 1 = 15$$

$$I = 15$$

Q22) A dataset has 3 attribute (A_1, A_2, A_3) each with 3 distinct values. What is the maximum Number of internal Nodes in decision tree built using all Attributes

Sol

3 attributes
Each attributes has
3 distinct value

*

Level 0: 1
Level 1: 3
Level 2: $3^2 = 9$

internal Nodes

$$I = 1 + 3 + 9$$

$$I = 13$$

max no. of leaves

$$L = 3^3 = 27$$

$$I = 13$$

Q23) Given a Node with 50 Ex (30+) 20-) split into two Nodes Left (20) Right (10 Pos & 15-). calculate Gain Ratio.

Sol

Node	(+)	(-)	Total
Parent	30	20	50
Left	20	5	25
Right	10	15	25

Entropy

$$H(S) = -P_+ \log_2 P_+ - P_- \log_2 P_-$$

$$H(S) = -0.6 \log_2(0.6) - 0.4 \log_2(0.4)$$

$$H(S) \approx 0.971$$

$$H(L) = -0.8 \log_2(0.8) - 0.2 \log_2(0.2)$$

$$H(L) \approx 0.722$$

Right child

$$H(R) = -0.4 \log_2(0.4) - 0.6 \log_2(0.6)$$

$$H(R) \approx 0.971$$

Information gain

$$IG = H(S) - \left[\frac{25}{50} H(L) + \frac{25}{50} H(R)\right]$$

$$IG = 0.971 - [0.5(0.722) + 0.5(0.971)]$$

$$IG \approx 0.1245$$

Split information $I = -\sum_i P_i \log_2 P_i$

$SI = -0.5 \log_2(0.5) - 0.5 \log_2(0.5)$

$SI = 1$

$GR = \frac{IG}{SI}$

$GR = \frac{0.1245}{1}$

$GR = 0.124$

Q24) In decision tree pruning using Reduced Error pruning, the validation accuracy improves after pruning a subtree.

Sol

Reduce Error pruning use validation data to decide whether a subtree should be remove If validation accuracy improve after pruning

Sol overfitting was reduced

Q25) A feature has 4 possible values and a split is performed on it The intrinsic value (IV) for the feature calculated.

Sol

$$IV = -\sum_{i=1}^{k} P_i \log_2 P_i$$

$P_i = 1/4$

$IV = -4(1/4 \log_2 1/4)$

$\log_2(1/4) = -2$

$IV = -4(1/4 * -2)$

$IV = 2$ bits